



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Class Activity – 24/07/2026

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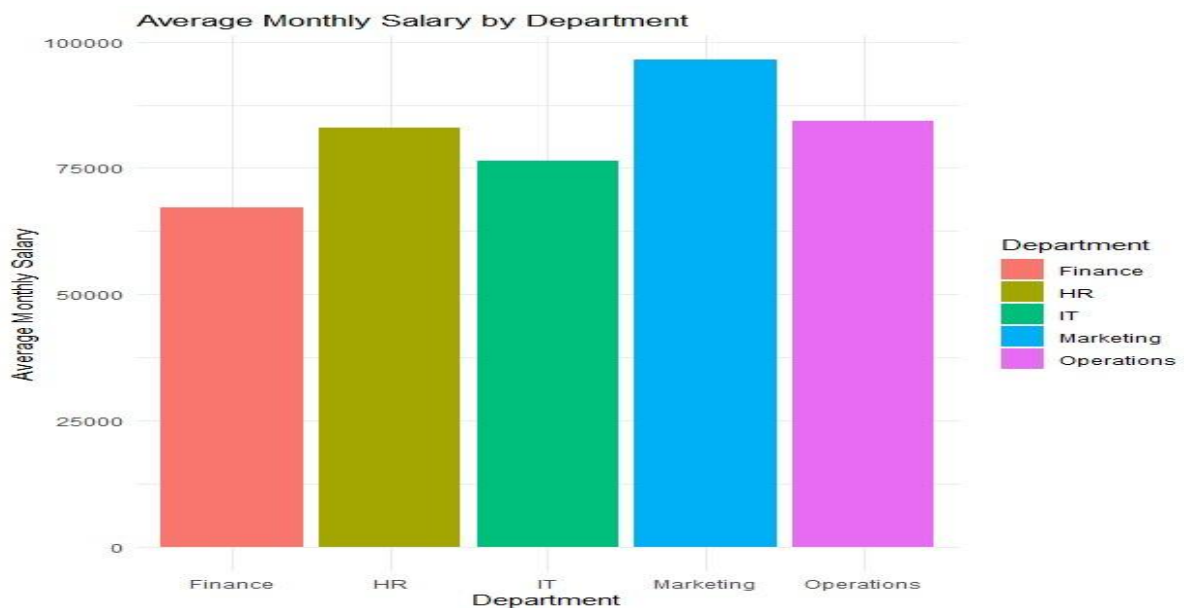
Reg No: 192424218

Employee Salary Visualizations

1.Bar chart

R Code:

```
bar_data <- aggregate(Monthly_Salary ~ Department, employee,
mean) ggplot(bar_data, aes(x = Department, y =
Monthly_Salary, fill = Department)) + geom_bar(stat =
"identity") + labs(title = "Average Monthly Salary by Department",
x = "Department", y = "Average Monthly Salary") +
theme_minimal() Output:
```

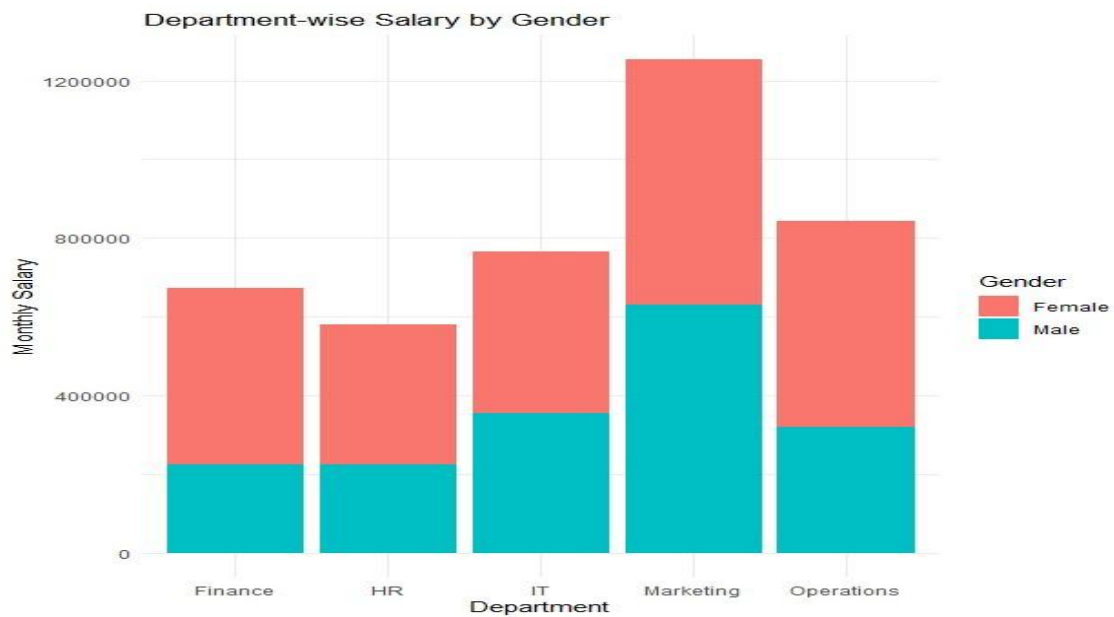


2.Stacked bar chart

R Code:

```
ggplot(employee, aes(x =
Department, y =
Monthly_Salary, fill =
Gender)) + geom_bar(stat =
"identity") +
```

```
labs(title = "Department-wise Salary by  
Gender",    x = "Department",    y =  
"Monthly Salary") + theme_minimal() Output:
```

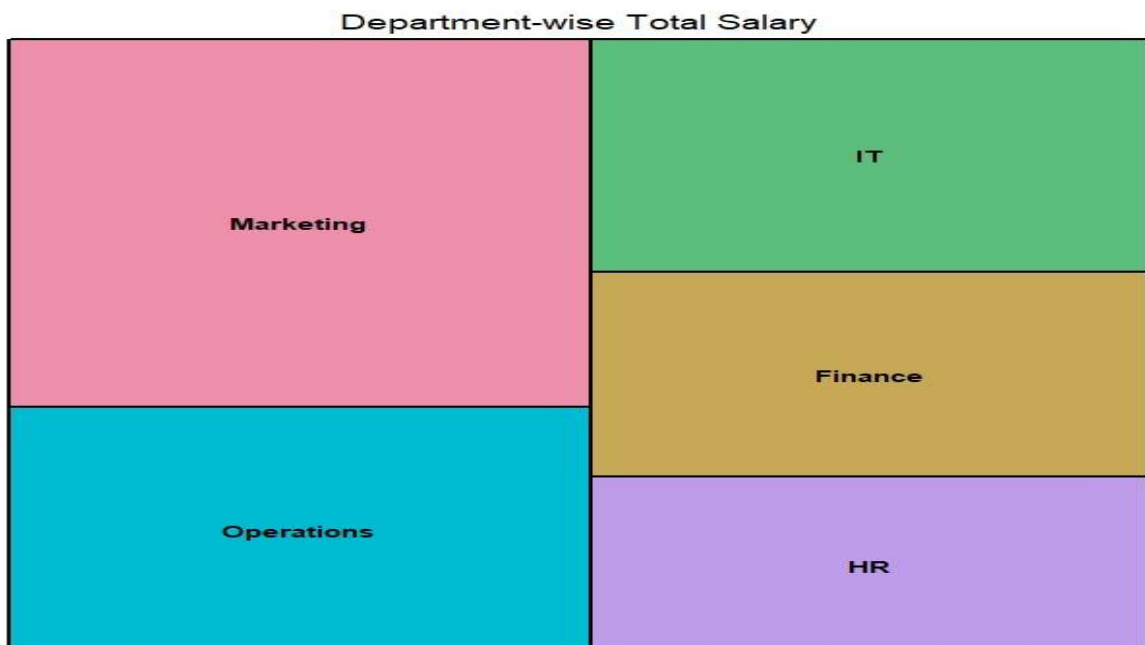


3.Tree Map

R Code:

```
tree_data <- aggregate(Monthly_Salary ~ Department, employee,  
sum) treemap(tree_data,    index = "Department",    vSize =  
"Monthly_Salary",    title = "Department-wise Total Salary")
```

Output:

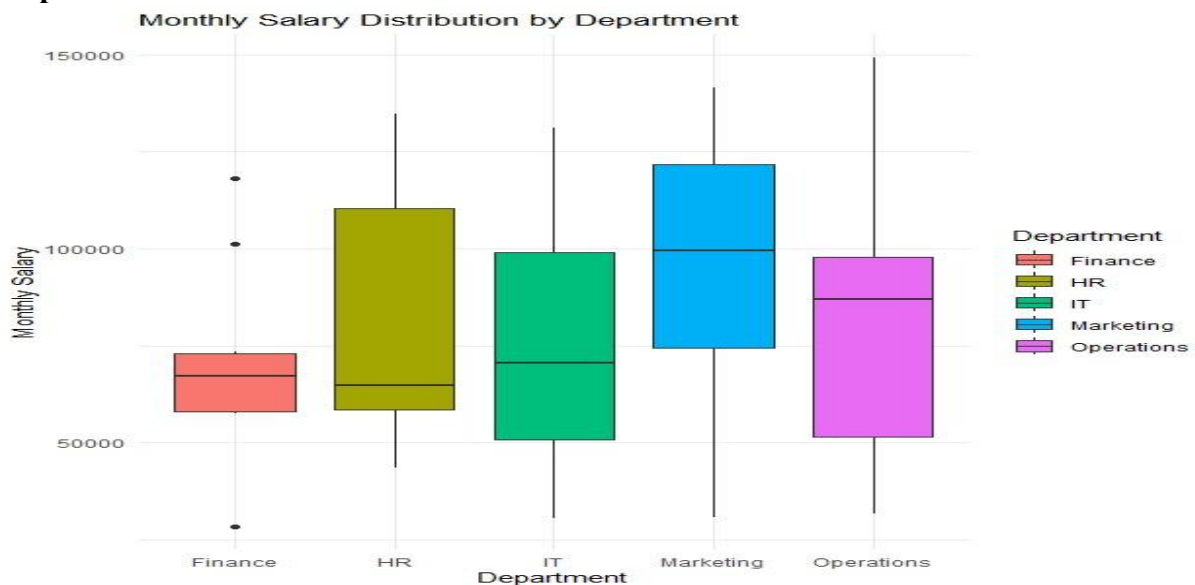


4.Box plot

R Code:

```
ggplot(employee, aes(x  
= Department, y =  
Monthly_Salary, fill  
= Department)) +  
geom_boxplot() +  
labs(title = "Monthly Salary Distribution by  
Department", x = "Department", y = "Monthly  
Salary") + theme_minimal()
```

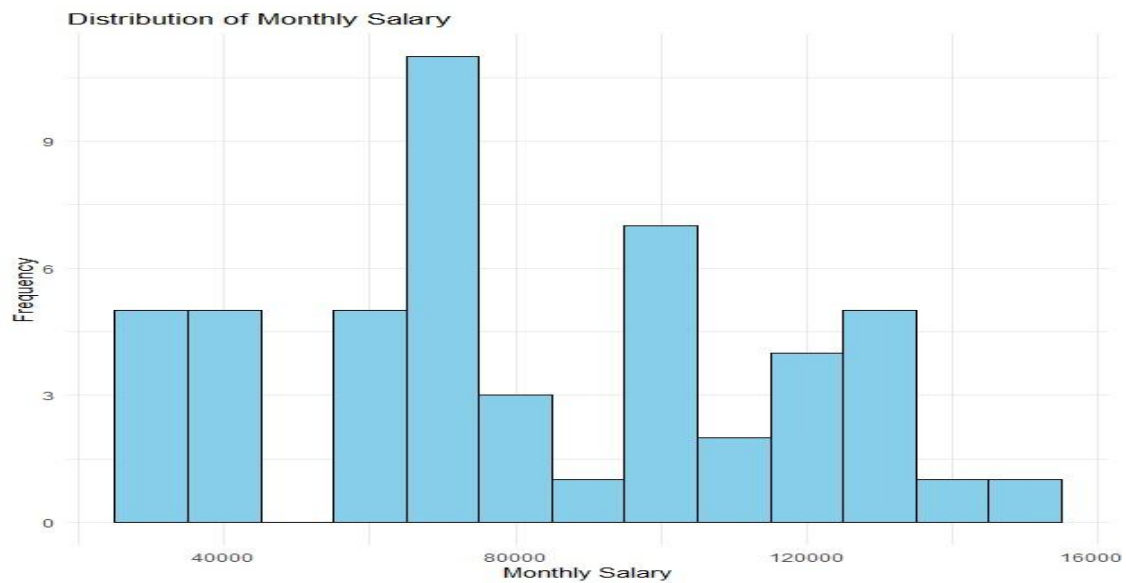
Output:



5.Histogram

R Code:

```
ggplot(employee,  
aes(x = Monthly_Salary)) +  
geom_histogram(binwidth = 10000,  
fill = "skyblue",  
color = "black") +  
labs(title = "Distribution of Monthly  
Salary", x = "Monthly Salary", y =  
"Frequency") + theme_minimal() Output:
```

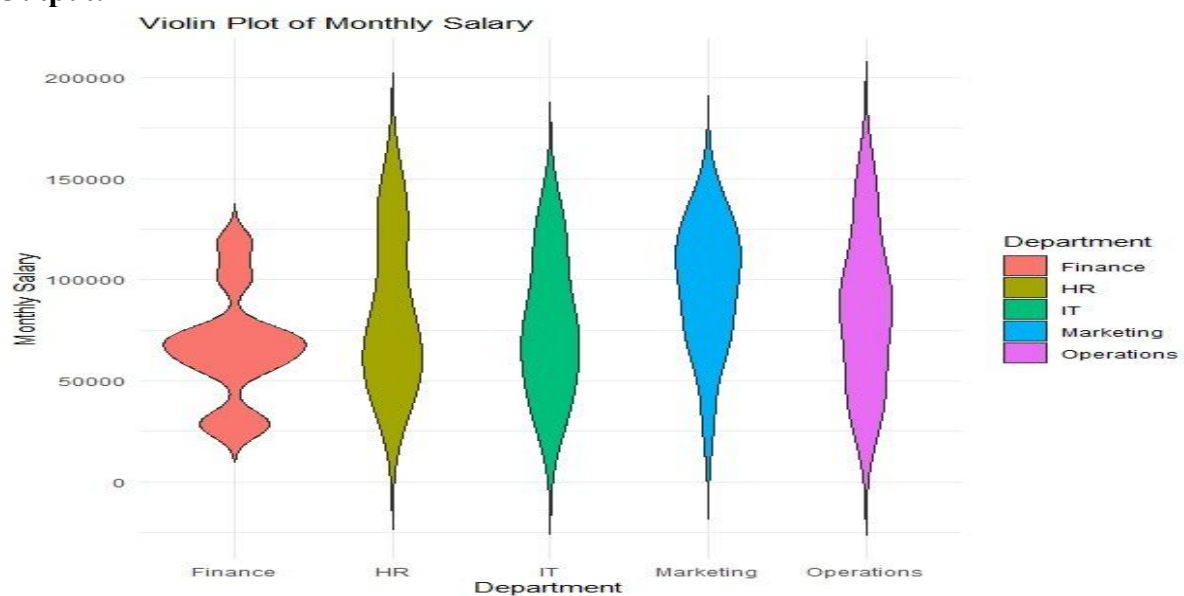


6. Violin plot

R Code:

```
ggplot(employee, aes(x = Department,
y = Monthly_Salary, fill =
Department)) + geom_violin(trim =
FALSE) + labs(title = "Violin Plot of
Monthly Salary", x = "Department",
y = "Monthly Salary") + theme_minimal()
```

Output:

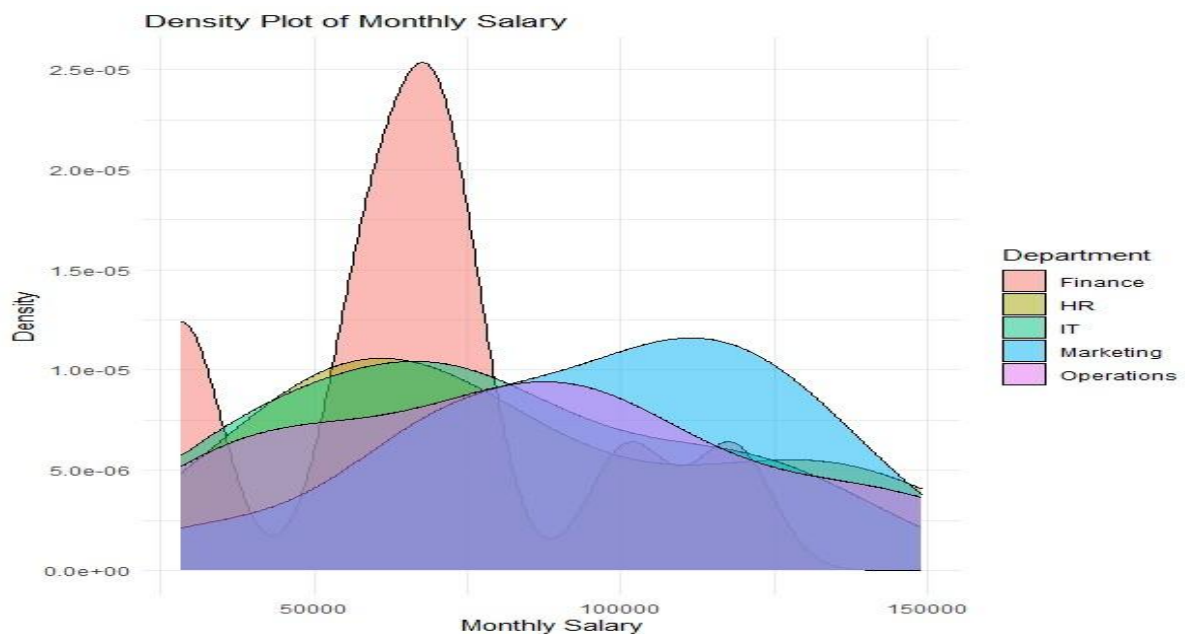


7. Density plot

R Code:

```
ggplot(employee, aes(x =
Monthly_Salary, fill =
Department)) +
geom_density(alpha = 0.5) +
labs(title = "Density Plot of Monthly Salary",
x = "Monthly Salary", y = "Density") +
theme_minimal()
```

Output:

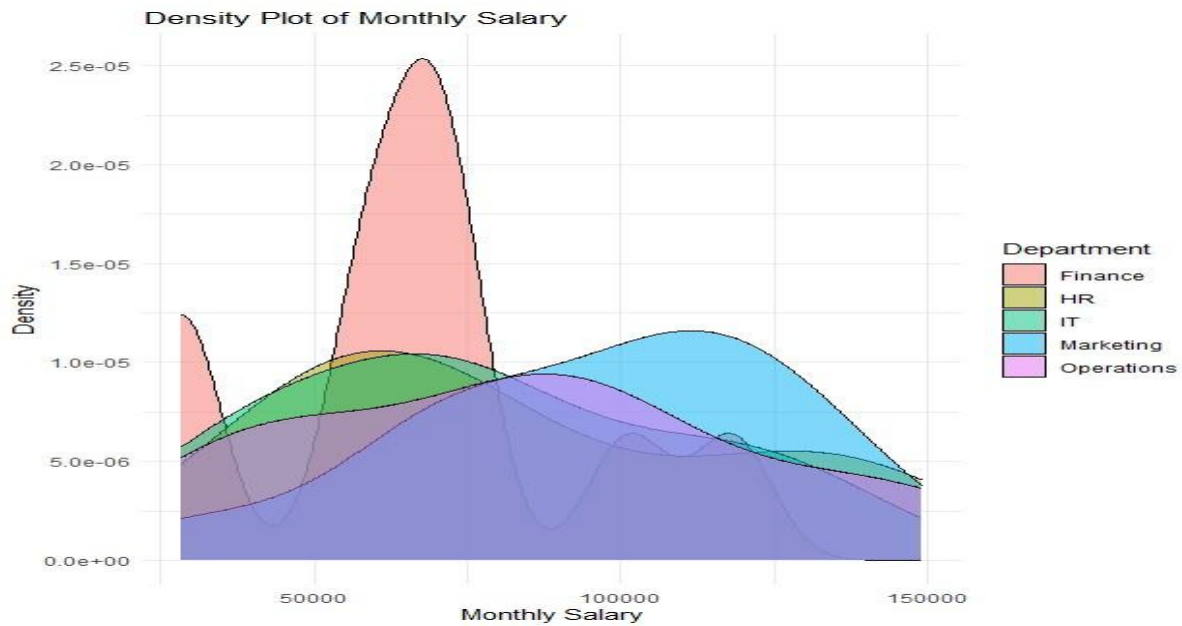


8.Scatter plot

R Code:

```
ggplot(employee,
aes(x = Experience_Years,
y = Monthly_Salary,
color = Department)) +
geom_point(size = 3) +
labs(title = "Experience vs Monthly
Salary", x = "Experience (Years)", y
= "Monthly Salary") + theme_minimal()
```

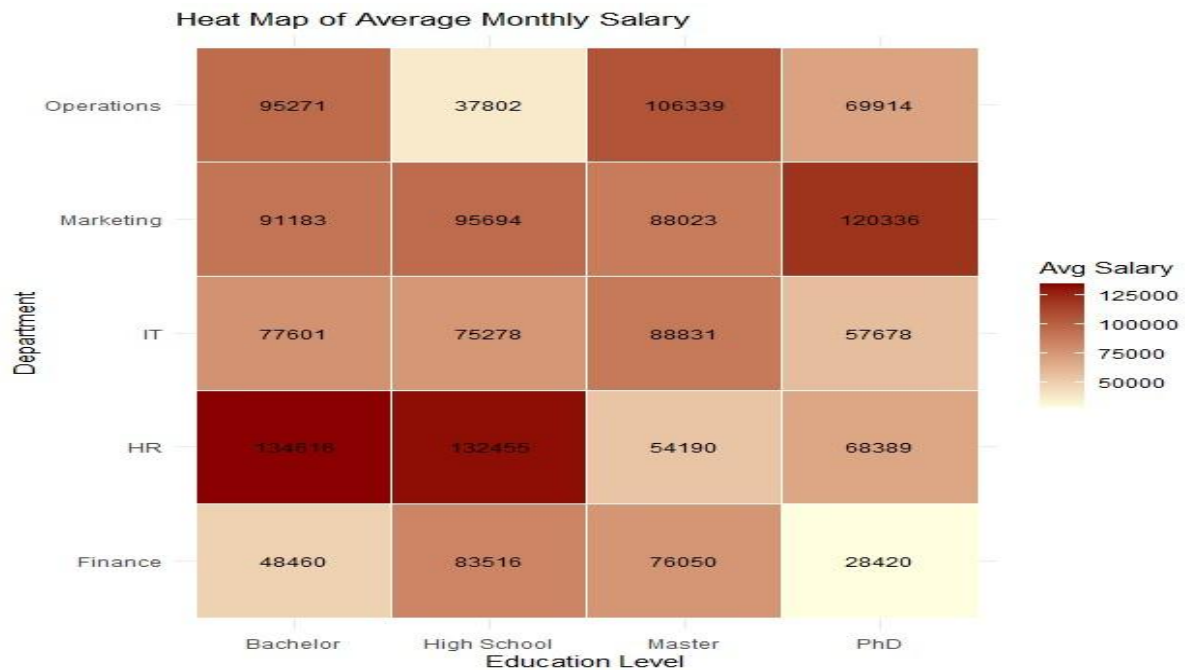
Output:



9.Heat map

R Code:

```
heat_data <- employee %>%
  group_by(Department, Education_Level) %>%
  summarise(Avg_Salary = mean(Monthly_Salary), .groups = "drop")
ggplot(heat_data, aes(x = Education_Level, y =
  Department, fill = Avg_Salary)) + geom_tile(color =
  "white") +
  geom_text(aes(label = round(Avg_Salary, 0)),
    color = "black", size = 3) +
  scale_fill_gradient(low = "lightyellow",
  high = "darkred") + labs(title = "Heat Map of
  Average Monthly Salary", x = "Education
  Level", y = "Department", fill = "Avg
  Salary") + theme_minimal() Output:
```



10. Bubble chart

R Code:

```
library(ggplot2) ggplot(employee, aes(x =
Experience_Years, y = Monthly_Salary,
size = Age, color = Department)) +
geom_point(alpha = 0.7) + labs(title = "Bubble
Chart of Employee Salary", x = "Experience
(Years)", y = "Monthly Salary", size =
"Age",
color = "Department")
+ theme_minimal()
```

Output:



11.Dot chart

R Code:

```
dot_data <- aggregate(Monthly_Salary ~ Department, employee,
mean) library(ggplot2) ggplot(dot_data, aes(x = Department,
y = Monthly_Salary, color = Department)) + geom_point(size
= 5) +
labs(title = "Average Monthly Salary by Department",
x = "Department", y = "Average Monthly Salary")
+ theme_minimal()
```

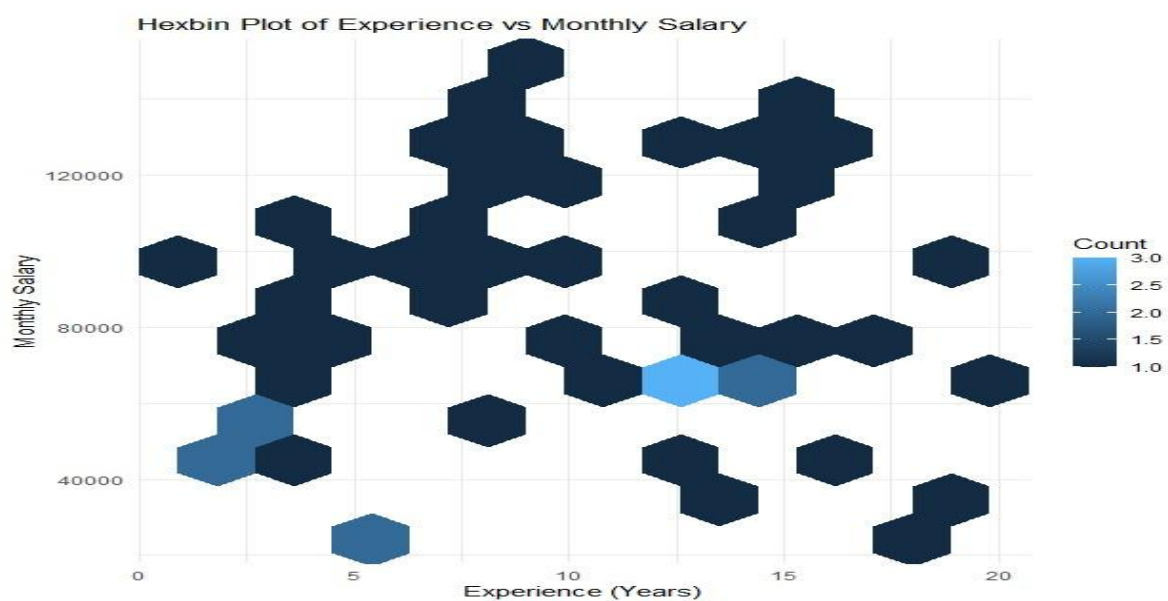
Output:



12. Bubble chart

R Code:

```
library(ggplot2)
library(hexbin)
ggplot(employee,
  aes(x = Experience_Years,
    y = Monthly_Salary)) +
  geom_hex(bins = 10) +
  labs(title = "Hexbin Plot of Experience vs Monthly
Salary", x = "Experience (Years)", y = "Monthly
Salary", fill = "Count") + theme_minimal() Output:
```



13. Line chart

R Code:

```
library(ggplot2)
```

```
line_data <- aggregate(Monthly_Salary ~ Experience_Years,
employee,
mean) ggplot(line_data,
aes(x = Experience_Years, y =
Monthly_Salary)) + geom_line(color =
"blue", linewidth = 1) + geom_point(color
= "red", size = 3) +
labs(title = "Average Monthly Salary by Experience",
x = "Experience (Years)", y = "Average Monthly
Salary") + theme_minimal() Output:
```

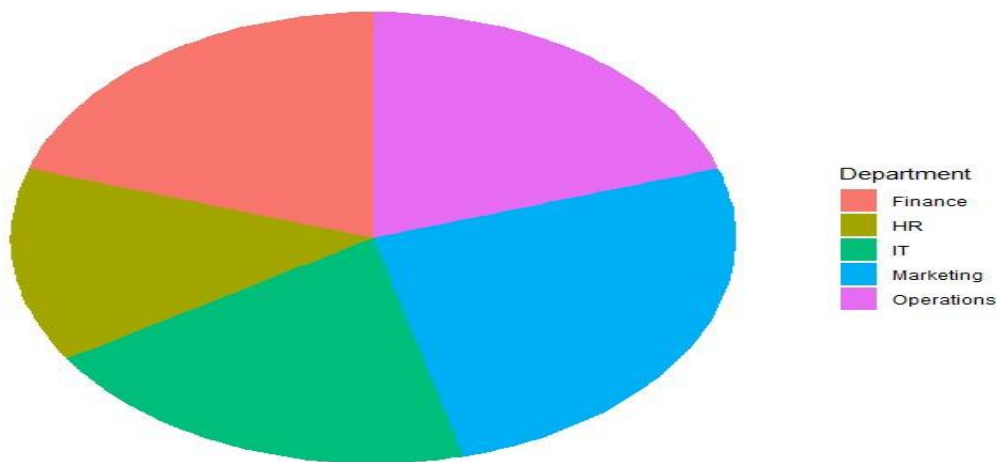


14. Pie chart

R Code:

```
library(ggplot2)
pie_data <-
as.data.frame(table(employee$Department))
colnames(pie_data) <- c("Department", "Count")
ggplot(pie_data, aes(x = "",
y = Count, fill =
Department)) + geom_bar(stat =
"identity", width = 1) +
coord_polar("y", start = 0) +
labs(title = "Employee Distribution by Department") +
theme_void() Output:
```

Employee Distribution by Department



15.Area chart

R Code:

```
library(ggplot2)
area_data <- aggregate(Monthly_Salary ~ Experience_Years,
employee,                mean)
ggplot(area_data, aes(x =
Experience_Years,        y =
Monthly_Salary)) + geom_area(fill =
"skyblue", alpha = 0.7) +
labs(title = "Area Chart of Average Salary by Experience",
x = "Experience (Years)", y = "Average Monthly
Salary") + theme_minimal()
```

Output:

